

ParkiGait

Parkinson's screening from a 30 second phone video — built for clinics with no neurologist. Research prototype, NOT a medical device.

The access problem

Parkinson's is the fastest growing neurological disorder in the world, and in most of the world the bottleneck is access: screening needs a specialist, and huge regions have almost none, so people are diagnosed late or never. The two signals a basic smartphone already records, a short walking video and a voice recording, may be enough to put the right people in front of the region's one specialist.

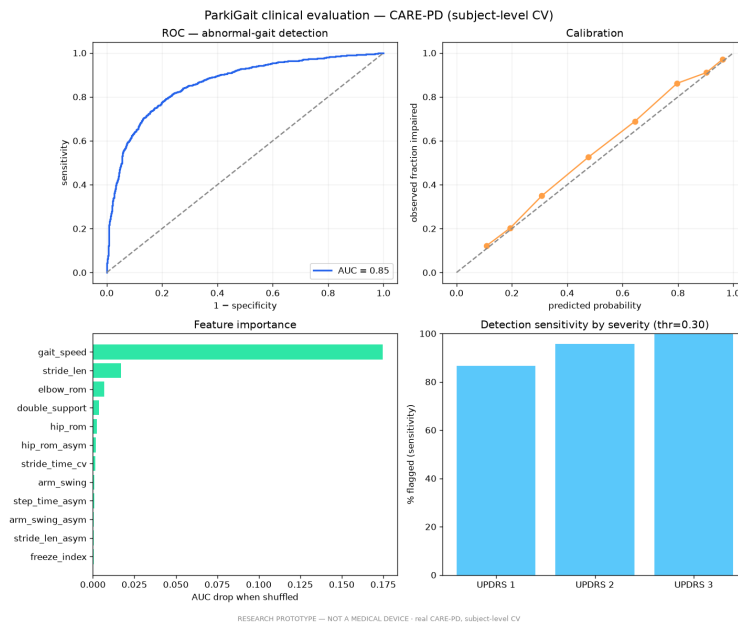
What it does

ParkiGait takes a short walking video and runs entirely on the phone or laptop: no cloud, no connectivity requirement, zero marginal cost per screen, and the patient video never leaves the device. It extracts a body skeleton, computes the gait features a clinician looks for (walking speed, stride length, arm swing, freezing, step to step variability), and flags possible parkinsonian gait for a clinician to review.

What the data says

Gait: trained and evaluated on the public CARE-PD dataset (110 patients with clinician MDS-UPDRS gait scores, ~2953 walks), strict subject level cross validation, 95% CIs by patient level bootstrap. Voice: companion open source screen, three independent public cohorts.

Task	Result
Detect impaired gait (screening) — AUC	0.86 to 0.87 (95% CI 0.82 to 0.90), permutation $p < 0.005$
External validation (a hospital never seen in training)	pooled AUC 0.77 (0.81 to 0.86 at the larger sites)
Severity staging (UPDRS-gait 0 to 3)	quadratic weighted kappa 0.62, 95% within one level
Screening operating point	sensitivity 0.90, specificity 0.59, PPV 0.75, NPV 0.81
Companion voice screen (3 independent public cohorts)	honest subject level AUC 0.83 to 0.88, permutation clean
Speed / memory / privacy	~27 ms per frame, under 400 MB, fully on-device (no cloud)



Why the numbers can be trusted

Subject level splits everywhere, enforced by a test that fails the build if a person crosses train and test. A label permutation control ships with every reported number. External validation on a hospital never seen in training. And the repository flags its own optimistic numbers instead of headlining them. Not a diagnosis, not a replacement for a clinician, not ready for any patient care decision.

The long run, and what I am seeking

The version that matters: a community health worker in a clinic with no neurologist records thirty seconds of walking and talking, and the region's one specialist sees a ranked list instead of a waiting room. I am seeking an honest read from people who move medical technology into low resource settings, and guidance on the responsible next step, for example a small ethics approved study comparing the tool's flag to clinician assessment on anonymized videos.

Parv Mehndiratta, Dougherty Valley High School | open source code and full honest documentation: github.com/0103-parv/parkinsons-detection | joints are a canonical-skeleton approximation; gait validated on one public dataset; see the model card and clinical evaluation in the repo.